8. Let $A = \{a, b, c\}$. What is the equivalence relation of smallest cardinality on A? What is the equivalence relation of largest cardinality on A?
9. In the set Z of integers, define mRn if $m - n$ is divisible by 7. Prove that R is an equivalence relation.

1.6 Functions

Suppose that a particle is moving in the space. We assume the physical particle as a point. As time varies, the particle changes its position. Mathematically at any time the point occupies a position in the three dimensional space $\mathbb{R}^3$. Let us assume that the time varies from 0 to 1. So the movement or *functioning* of the particle decides the position of the particle at any given time t between 0 and 1. In other words, for each $t \in [0, 1]$, the functioning of the particle gives a point in $\mathbb{R}^3$. Let us denote the position of the particle at time t as $f(t)$.

Let us see another simple example. We know that the equation $2x - y = 0$ describes a straight line. Here whenever x assumes a value, y assumes some value accordingly. The movement or *functioning* of y is decided by that of x. Let us denote y by $f(x)$. We may see many situation like this in nature. In the study of natural phenomena, we find that it is necessary to consider the variation of one quantity depending on the variation of another.

The relation of the time and the position of the particle, the relation of a point in the x-axis to a point in the y-axis and many more such relations are studied for a very long period in the name *function*. Before Cantor, the term function is defined as a rule which associates a variable with another variable. After the development of the concept of sets, a function is defined as a rule that associates for every element in a set A, a unique element in a set B. However the terms *rule* and *associate* are not properly defined mathematical terminologies. In modern mathematics every term we use has to be defined properly. So a definition for function is given using relations.

Suppose that we want to discuss a test written by a set of students. We shall see this as a relation.

Let A be the set of students appeared for an examination and let $B = \{0, 1, 2, 3, \ldots 100\}$ be the set of possible marks. We define a relation R as follows:

A student a is related to a mark b if a got b marks in the test.

We observe the following from this example:

- Every student got a mark. In other words, for every $a \in A$, there is an element $b \in B$ such that $(a, b) \in R$.
- A student cannot get two different marks in any test. In other words, for every $a \in A$, there is definitely only one $b \in B$ such that $(a, b) \in R$. This can be restated in a different way: If $(a, b), (a, c) \in R$ then $b = c$.

Relations having the above two properties form a very important class of relations, called functions. Let us now have a rigorous definition of a function through relations.

Definition 1.6

Let A and B be two sets. A relation f from A to B, a subset of $A \times B$, is called a **function** from A to B if it satisfies the following:

(i) for all $a \in A$, there is an element $b \in B$ such that $(a, b) \in f$.
(ii) if $(a, b) \in f$ and $(a, c) \in f$ then $b = c$.

That is, a function is a relation in which each element in the domain is mapped to exactly one element in the range.

A is called the ***domain*** of f and B is called the ***co-domain*** of f. If (a, b) is in f, then we write $f(a) = b$; the element b is called the ***image*** of a and the element a is called a ***pre-image*** of b and $f(a)$ is known as the value of f at a. The set $\{b : (a, b) \in f \text{ for some } a \in A\}$ is called the ***range*** of the function. If B is a subset of $\mathbb{R}$, then we say that the function is a ***real-valued function***.